

CE802 - QUESTIONS

This is a pdf version of a Moodle-based exam.

This exam is Open Book (Restricted), students may use the following in the exam: Formulary (available at the end of the exam).

Candidates are permitted to use: Calculator — Casio FX-83GT or Casio FX-85GT Series ONLY (CW, X, Plus)

Candidates must answer ALL QUESTIONS.

The percentages shown in brackets at the end of each question provide an indication of the proportion of the total marks for the PAPER which will be allocated.

Questions may be of type:

- **MCQ:** The student needs to select the only correct answer among a set of choices.
- **Cloze:** These are questions with gaps (highlighted in yellow) where the student is required to select an answer from a list of choices (the question shows the list of choices separated by “/”) or enter a numerical value (with the placeholder shown as “[?]”).
- **Open:** These are open questions where the student is asked to enter a response in a text box. Word limits are indicated in the question.

1. Question

Suppose you have an incremental 1 nearest neighbour learning system that uses the number of identical attributes as its similarity measure. The system retains a new example if it satisfies any of the following conditions (tested in this order):

- 1 It has zero similarity to saved examples;
- 2 It has two or more equally similar nearest neighbours of different classes;
- 3 It would have been misclassified using single nearest neighbour.

Examples that do not satisfy any of these conditions are not retained. The data set given in the table below is presented to this system only once and in the order that it appears in the table.

Size	Shape	Colour	Reliability
medium	sphere	red	Poor
medium	pyramid	brown	Poor
small	sphere	yellow	Good
small	sphere	red	Good
medium	sphere	yellow	Poor
large	cube	brown	Good
small	sphere	brown	Poor
large	cube	red	Good

Explain which examples will be retained and why.

medium,sphere,red : Poor [?Not retained / Retained because of condition 1?]

medium,pyramid,brown : Poor [?Not retained / Retained because of condition 1 / Retained because of condition 2 / Retained because of condition 3?]

small,sphere,yellow : Good [?Not retained / Retained because of condition 1 / Retained because of condition 2 / Retained because of condition 3?]

small,sphere,red : Good [?Not retained / Retained because of condition 1 / Retained because of condition 2 / Retained because of condition 3?]

medium,sphere,yellow : Poor [?Not retained / Retained because of condition 1 / Retained because of condition 2 / Retained because of condition 3?]

large,cube,brown : Good [?Not retained / Retained because of condition 1 / Retained because of condition 2 / Retained because of condition 3?]

small,sphere,brown : Poor [?Not retained / Retained because of condition 1 / Retained because of condition 2 / Retained because of condition 3?]

large,cube,red : Good [?Not retained / Retained because of condition 1 / Retained because of condition 2 / Retained because of condition 3?]

2. Question

In the context of supervised learning, a large variance (representing the quantity of modelled noise) is associated with:

- (a) overfitting and can be reduced by reducing the complexity of the hypothesis class
- (b) underfitting and can be reduced by reducing the complexity of the hypothesis class
- (c) underfitting and can be reduced through feature selection
- (d) overfitting and can be reduced by reducing the number of examples in the training set

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3. Question

Testing a classifier's performance using a 10-fold cross-validation scheme produced the following confusion matrix:

Actual	Predicted	
	Yes	No
Yes	45	8
No	5	64

Please compute the following performance measures (as ratios $\in [0, 1]$, not as percentages):

- Estimated *prior probability* of class Yes [?0.678 / 0.849 / 0.9 / 0.893 / 0.241 / 0.113 / 0.473 / 0.434 / 0.777 / 0.0761?]
- *Recall* (also known as *true positive rate* or *sensitivity*) [?0.678 / 0.849 / 0.9 / 0.893 / 0.241 / 0.113 / 0.473 / 0.434 / 0.777 / 0.0761?]
- *Precision* (also called *positive predictive value*) [?0.678 / 0.849 / 0.9 / 0.893 / 0.241 / 0.113 / 0.473 / 0.434 / 0.777 / 0.0761?]
- Classification *accuracy* [?0.678 / 0.849 / 0.9 / 0.893 / 0.241 / 0.113 / 0.473 / 0.434 / 0.777 / 0.0761?]

[6]

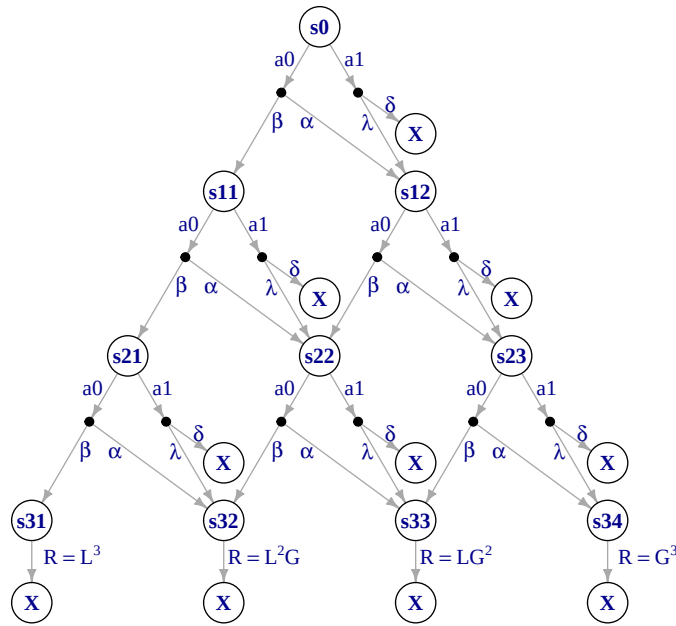
4. Question

You have a resource that you want to manage over a 3-year period. You start with a small amount of the resource, and each year you must decide how to allocate it, according to the following two options:

- **Option 0:** Allocate the resource to a lower-risk activity that may result in a large increase (by a factor $G = 7$) of the current amount with probability $\alpha = 1/7$ or a reduction (by a factor $L = 1/6$) of the current amount with probability $\beta = 1 - \alpha = 6/7$.
- **Option 1:** Allocate the resource to a higher-risk activity that may result in the large increase (still by a factor G) with probability $\lambda = 1/4$ (higher than for Option 0), but may also result in a complete loss of the resource with probability $\delta = 1 - \lambda = 3/4$. If you lose the resource, you are out of business and cannot continue.

The outcome of each year's allocation is independent of the previous ones. Your goal is to maximize the amount of the resource after 3 years.

We can model this scenario through a Markov Decision Process (MDP) as represented in the diagram below.



The $\langle S, A, T, R \rangle$ is as follows:

- **States:** The state of the MDP is the amount of the resource available at the beginning of each year. While in principle the state is continuous, if we assume a nominal initial amount of the resource (e.g. 1) then only a discrete number of states becomes feasible. At time $t = 0$ there is only one possible state s_0 (where the amount for the resource is 1). If at this or any later time, the agent loses everything, they enter state

X and the game ends. Assuming a complete loss has not occurred, there are only two possible states at time $t = 1$: s_{11} where L amount of resource is available and s_{12} where G amount of resource is available. Three states are possible at time $t = 2$, with amounts L^2 , LG , and G^2 , and four at $t = 3$, with amounts L^3 , L^2G , LG^2 and G^3 . Only at this time, the agent can “cash” their earnings (in the form of a reward) and the game ends.

- **Actions:** In each year $t \in \{0, 1, 2\}$ the agent can take either action a_0 corresponding to “Option 0” or action a_1 corresponding to “Option 1”. At time $t = 3$ the only action possible is to “cash” the earnings.
- **Transition probabilities:** The transition probabilities are represented in the graph. For example:
 - $T(s_{22}|s_{12}, a_0) = [2/63 / 7/36 / 343/64 / 49/96 / 7/2 / 6/7 / 1/3 / 7/144 / 1/216 / 1/7?]$
 - $T(s_{23}|s_{12}, a_0) = [2/63 / 7/36 / 343/64 / 49/96 / 7/2 / 6/7 / 1/3 / 7/144 / 1/216 / 1/7?]$
- **Rewards:** Since the only goal is to maximize the amount of the resource after 3 years, the only actions leading to a reward are the “cash” actions out of $s_{31} \dots s_{34}$. These rewards correspond to: $R(s_{31}, \text{cash}) = L^3$, $R(s_{32}, \text{cash}) = L^2G$, $R(s_{33}, \text{cash}) = LG^2$ and $R(s_{34}, \text{cash}) = G^3$.

Since the goal is to maximize the amount of the resource after 3 years, we can use a discount factor of 1, which means that the rewards are not discounted over time.

In order to decide the optimal policy, you need to compute the state-action value function $Q(s, a)$. You can use the “value iteration” procedure that alternates between updating the Q s and the V s. Alternatively, you can apply “Bellman’s optimality equation” directly, provided that you compute the Q s in a suitable order.

After the procedure, please report some of the computed Q values as indicated below:

- $Q^*(s_0, a_0) = [2/63 / 7/36 / 343/64 / 49/96 / 7/2 / 6/7 / 1/3 / 7/144 / 1/216 / 1/7?]$
- $Q^*(s_0, a_1) = [2/63 / 7/36 / 343/64 / 49/96 / 7/2 / 6/7 / 1/3 / 7/144 / 1/216 / 1/7?]$
- $Q^*(s_{11}, a_0) = [2/63 / 7/36 / 343/64 / 49/96 / 7/2 / 6/7 / 1/3 / 7/144 / 1/216 / 1/7?]$
- $Q^*(s_{11}, a_1) = [2/63 / 7/36 / 343/64 / 49/96 / 7/2 / 6/7 / 1/3 / 7/144 / 1/216 / 1/7?]$
- $Q^*(s_{21}, a_0) = [2/63 / 7/36 / 343/64 / 49/96 / 7/2 / 6/7 / 1/3 / 7/144 / 1/216 / 1/7?]$
- $Q^*(s_{21}, a_1) = [2/63 / 7/36 / 343/64 / 49/96 / 7/2 / 6/7 / 1/3 / 7/144 / 1/216 / 1/7?]$
- $Q^*(s_{31}, \text{cash}) = [2/63 / 7/36 / 343/64 / 49/96 / 7/2 / 6/7 / 1/3 / 7/144 / 1/216 / 1/7?]$
- $Q^*(s_{32}, \text{cash}) = [2/63 / 7/36 / 343/64 / 49/96 / 7/2 / 6/7 / 1/3 / 7/144 / 1/216 / 1/7?]$
- $R(s_{33}, \text{cash}) = 49/6$
- $R(s_{34}, \text{cash}) = 343$

Finally, based on the results above, the optimal policy is: $\pi(s) = [a_0 \text{ for all } s \text{ at time } 0, 1 \text{ and } 2 / a_1 \text{ for all } s \text{ at time } 0, 1 \text{ and } 2 / a_1 \text{ at time } 0, \text{ then } a_0 \text{ at time } 1 \text{ and } 2 / a_0 \text{ at time } 0 \text{ and } 1, \text{ then } a_1 \text{ at time } 2?]$.

5. Question

Consider the previous question about managing a resource over three years. Please answer the following questions about the optimal policy and its implications:

- (a) You have found that the optimal policy is to always choose a given action, despite the fact that it has a high probability of resulting in a complete loss of the resource. How does this policy align with your intuition about risk management? Shouldn't a more conservative approach be preferred, especially since the expected value of both actions is greater than one?
- (b) Consider the long-term implications of the optimal policy. If the duration of the experiment were to be extended to k years (instead of three), what would be the probability of cashing a reward (or equivalently the probability of not losing everything) under the optimal policy? What would that probability be if the experiment were to be extended indefinitely (i.e. $k \rightarrow \infty$)? Why do you think the Q^* -based optimal policy leads to this result?
- (c) Suppose you were to modify the problem to include a "survival" constraint, where the goal is not only to maximize the amount of the resource but also to ensure that the resource is not completely lost. How would you modify the reward function to incorporate this constraint, and what implications would this have for the optimal policy?

Note: Keep your response concise, ensuring it does not exceed 250 words.

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6. Question

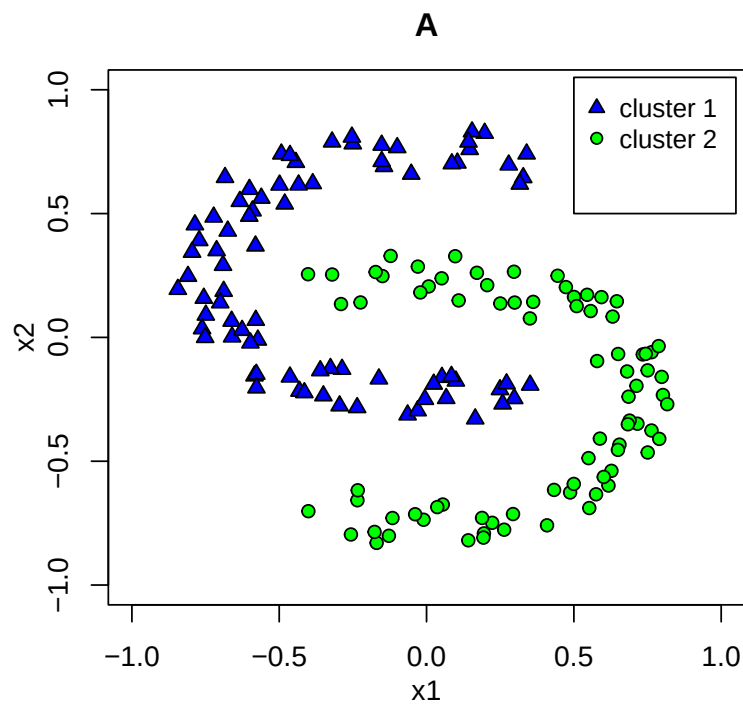
Use the k-means algorithm with Euclidean distance to cluster the following 6 examples into 2 clusters: $a=(3, -5)$, $b=(-3, -1)$, $c=(-2, 2)$, $d=(0, 1)$, $e=(2, 2)$, $f=(5, -5)$. Suppose that the initial centroids are: $c_1 = (-3, -1)$, $c_2 = (2, 2)$. Run the k-means algorithm for 3 iterations or until convergence, whichever comes first. For each iteration show the cluster assignments (which instances belong to each cluster) and the new centres of the clusters (please use at least three significant digits).

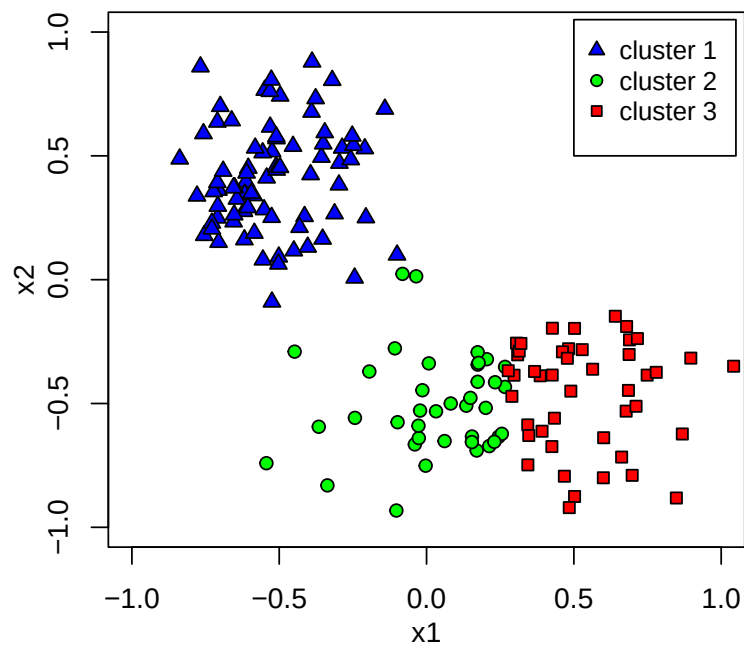
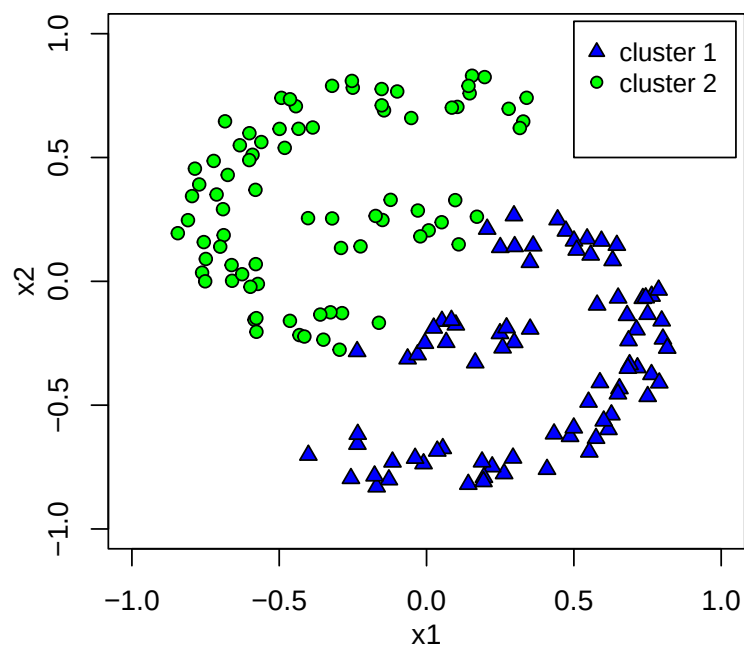
- At iteration 1 the cluster memberships are updated as $[a, b, c, d, e / a, c, d, e / a, b, d / a, d / b, c, d, e, f / b, d / a, e, f / a, c, d, e, f / b / b, c, ?]$ for cluster 1 and $[a, c, d, e, f / a, c, e, f / c, e, f / a, d, e, f / b / a / b, f / f / b, c, e, f / b, c, d, ?]$ for cluster 2; then the centroids are updated as follows:
 $c_1 = ([?], [?],)$
 $c_2 = ([?], [?],)$
- At iteration 2 the cluster memberships are updated as $[b, d / b, f / b, e, f / b, c, d / e / a, b / c, f / b, c, e / b, c, f / a, b, e, ?]$ for cluster 1 and $[a, b, c, d, f / a, d, f / a, c, d, e / a, c, e, f / c, d, f / a, d, e / a, b, d, e / c, d, e, f / a, c, d / a, e, f, ?]$ for cluster 2; then the centroids are updated as follows:
 $c_1 = ([?], [?],)$
 $c_2 = ([?], [?],)$
- At iteration 3 the cluster memberships are updated as $[b, c, f / b, c, d, e / b, f / a, b, c, d / a, d / b, c, e / a, b / a, c, d, e, f / c, e / b, d, e, ?]$ for cluster 1 and $[e, f / a, b, d, f / c, d, e, f / a, c, f / b / a, d, e / a, f / a, c, d, e / b, c, e, f / a, d, f, ?]$ for cluster 2; then the centroids are updated as follows:
 $c_1 = ([?], [?],)$
 $c_2 = ([?], [?],)$

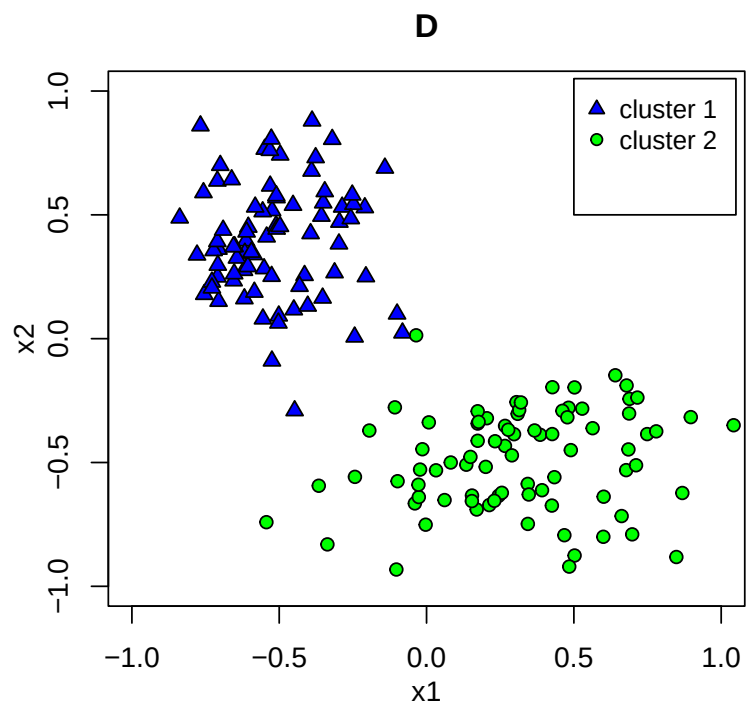
[18]

7. Question

Identify which one of the following clusterings was NOT obtained using k-means.
Briefly justify your answer using no more than 50 words.



B**C**



8. Question

Consider the following Python code snippet used for machine learning:

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.feature_selection import SelectKBest
from sklearn.model_selection import cross_val_score

# Load dataset and create X and y
df = pd.read_csv('data.csv')
X = df.drop('target', axis=1) # all columns except target
y = df['target']

# Feature selection
selector = SelectKBest(k=10)
X_selected = selector.fit_transform(X, y)

# Fit in cross-validation using previously selected features
model = DecisionTreeClassifier(criterion='entropy', max_depth=9)
scores = cross_val_score(model, X_selected, y, cv=5, scoring='accuracy')

# Print average cross-validation accuracy
print('Cross-Validation_Accuracy:', scores.mean())
```

Carefully analyze the code and identify potential issues that may affect the reliability of the model training process.

Your answer should cover the following points:

- Identify the main potential issue that might affect the reliability of the model training process.
- Explain the implications of the identified issue.
- Propose strategies or approaches to address the identified issue and provide a detailed rationale for the proposed solution.

Note: Keep your response concise, ensuring it does not exceed 250 words.

[15]

Machine Learning Formulary

General

- Empirical risk minimisation: $\hat{\theta} \in \arg \min_{\theta} \frac{1}{N} \sum_{t=1}^N L(r^{(t)}, h(x^{(t)}|\theta))$

Decision Trees

- Entropy/Information: $I(S) = -\sum_{i=1}^c p_i \log_2(p_i)$ (note that: $\log_2(x) = \ln(x)/\ln(2)$)

Naive Bayes

- Conditional probability: $P(A|B) := \frac{P(A \wedge B)}{P(B)}$
- Conditional probability (2+ events): $P(A|B \wedge C) = \frac{P(A \wedge B|C)}{P(B|C)} = \frac{P(A \wedge B \wedge C)}{P(B \wedge C)}$
- Bayes' Theorem: $P(A|B) = \frac{P(B|A) P(A)}{P(B)}$
- M-estimate of probability (p prior, m pseudo-observations): $\hat{P}(C) = \frac{n_C + mp}{n + m}$
- MAP hypothesis: $\arg \max_{\{C_1, \dots, C_M\}} \frac{P(C_k) \cdot P(A_1, \dots, A_N|C_k)}{P(A_1, \dots, A_N)}$

k-Nearest Neighbors (k-NN)

- Euclidean Distance: $d(x, x') = \sqrt{\sum_{i=1}^n (x_i - x'_i)^2}$
- Manhattan Distance: $d(x, x') = \sum_{i=1}^n |x_i - x'_i|$

Support Vector Machines

- Linear SVM Classifier:

$$\min_{w, b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n L_{\text{hinge}}(r^{(i)}, w^T x^{(i)} - b)$$

- Margin: $\frac{2}{\|w\|}$
- Hinge Loss: $L_{\text{hinge}}(r, y) = \max(0, 1 - ry)$
- Inference (with Kernels): $h(x) = \left[\sum_{s \in \mathcal{S}} \alpha_s r^{(s)} K(x^{(s)}, x) \right] - b$
- Gaussian radial basis function kernel: $K(x, x') = \exp(-\gamma \|x - x'\|^2)$

Classification Performance Metrics

- Precision or Positive Predictive Value: $\frac{TP}{TP+FP}$
- Recall or Sensitivity: $\frac{TP}{TP+FN}$
- Specificity: $\frac{TN}{TN+FP}$
- Accuracy: $\frac{TP+TN}{TP+FP+TN+FN}$

Regression

- Linear Regression Model: $r = h(x|\theta) + \epsilon = \theta_0 + \theta_1 x_1 + \dots + \theta_p x_p + \epsilon$
- Mean squared error: $\text{MSE} := \frac{1}{N} \sum_{i=1}^N (r^{(i)} - h(x^{(i)}|\theta))^2$
- Coefficient of determination: $R^2 := 1 - \frac{\sum_i (r^{(i)} - h(x^{(i)}|\theta))^2}{\sum_i (r^{(i)} - \bar{r})^2}$

Clustering

- k-Means Clustering – cluster centres: $c_k = \frac{1}{|C_k|} \sum_{x \in C_k} x$
- Hierarchical agglomerative clustering:

$$\text{Single Linkage (Minimum Linkage): } \text{dist}(C_j, C_k) = \min_{x \in C_j, x' \in C_k} \text{dist}(x, x')$$

$$\text{Complete Linkage (Maximum Linkage): } \text{dist}(C_j, C_k) = \max_{x \in C_j, x' \in C_k} \text{dist}(x, x')$$

- Single-cluster cohesion: $\text{Coh}(C_k) = \sum_{x \in C_k} \|x - c_k\|^2$
- Overall cohesion: $\text{Coh} = \sum_k \text{Coh}(C_k) = \sum_k \sum_{x \in C_k} \|x - c_k\|^2$
- Total variability (no clustering): $\text{Coh}_0 = \sum_x \|x - \bar{x}\|^2$

Reinforcement learning

- State transition probability: $T(s'|s, a) := \Pr(s_{t+1}=s' \mid s_t=s, a_t=a)$
- Expected reward: $R(s, a) := \mathbb{E}[r_{t+1} \mid s_t=s, a_t=a]$
- State-value function:

$$V^\pi(s) := \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t=s \right]$$

- Action-value function (or Q-function):

$$Q^\pi(s, a) := \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t=s, a_t=a \right]$$

- Bellman's Optimality Equation:

$$V^*(s) = \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} T(s'|s, a) V^*(s') \right]$$
$$Q^*(s, a) = R(s, a) + \gamma \sum_{s' \in S} T(s'|s, a) \max_{a' \in A} Q^*(s', a')$$

- Value iteration (batch version) alternates between:

$$\begin{aligned} \text{Update all Qs: } Q(s, a) &\leftarrow R(s, a) + \gamma \sum_{s' \in S} T(s'|s, a) V(s') \\ \text{Update all Vs: } V(s) &\leftarrow \max_{a \in A} Q(s, a) \end{aligned}$$

- Deterministic value iteration:

$$V^*(s) = \max_{a \in A} [R(s, a) + \gamma V^*(s')]$$

- Deterministic Q-learning:

$$Q^*(s_t, a_t) = r_{t+1} + \gamma \max_{a \in A} Q^*(s_{t+1}, a)$$